

Track A Lab Guide

Coin Rush: A Mobile Game in 9 Exercises with Unity CLI

Prepared for Panteon Games — Codeo · August 2026 · Engineering + DevOps · 13:30 – 15:00

Today you build a real mini game end to end in the coin-rush-workshop project: a robot character, 8 coins, a patrolling mine, score, win/lose, a magnet power-up and an iOS build. In this compressed single-trainer run the core exercises (Ex01, Ex02, Ex06, Ex07) are hands-on with verification from the terminal; Ex03–Ex05 and Ex09 are fast instructor demos, and Ex08 is briefed for you to complete after the workshop. Each exercise's detailed steps and copy-ready prompts live in the README file inside its own folder — this guide is your roadmap for the day.

THE SYNC RULE — NO ONE FALLS BEHIND

Every ExNN folder under Assets/Workshop contains a Start scene equal to the finished state of the previous exercise. At the start of each exercise EVERYONE opens that exercise's Start scene: whether you finished, got halfway, or missed it entirely, you begin from the same point. Nobody restores anything; the next folder is also the previous exercise's solution.

ENVIRONMENT CHECK (10 MIN)

- The coin-rush-workshop project is open on your machine; the Unity editor is ready.
- In the terminal, `unity status` shows the connected editor as "ready".
- A `claude` session opens; you have read the CLAUDE.md contract at the project root.
- Under Assets/Workshop you can see the Ex01...Ex09 and _Complete folders.
- You have opened _Complete/CoinRush.unity and played the finished game once (know the goal).

THE NINE EXERCISES

1. Ex01_SceneSetup — Meeting the CLI (13:40, hands-on)

`unity status`, the command list; create the ground and the player from the terminal with `eval` commands — no mouse. Verify: `eval 'GameObject.Find("Player") != null' » True`.

2. Ex02_Movement — The agent writes the first script (hands-on)

Using the prompt in the README, have Claude Code write `PlayerController` in plan mode. If the plan reaches outside the Ex02 folder, reject it. Verify: WASD and click-to-move work in play mode.

3. Ex03_Coins — The coin prefab and live queries (instructor demo)

Have the agent build the spinning coin prefab and the ring of 8. Verify: count the remaining coins from the terminal (`eval '...Coins...childCount'`) — this is the core muscle of bug hunting.

4. Ex04_Score — Score, winning, and multi-file work (instructor demo)

Four parts in one request: `ScoreManager`, `GameManager`, the `CoinPickup` change, UI. Verify: YOU WIN appears at 8/8; `eval 'Time.timeScale' » 0`.

5. Ex05_Hazard — The mine and iteration (instructor demo)

Have it build the patrol + lose flow; then change speed/route with at least one follow-up prompt. The first prompt is not the final behavior. Verify: touching the mine shows GAME OVER, RESTART works.

6. Ex06_BugHunt — BUG-201 (14:25, hands-on)

The scene ships with a QA report: some coins award 2 points. The rule: no fix until the root cause is proven. If the agent patches the symptom (a cap, a flag), reject the plan. Proof: `eval` shows 2 colliders on the coin; after the fix, 1 — and a full run ends 8/8.

7. Ex07_CliCommand — workshop_validate (14:45, hands-on)

Your own CLI command: a level validator that audits the scene ([`CliCommand`]). Verify: `unity command workshop_validate` is green; move the player to `x=50` and it turns red.

8. Ex08_Magnet — FEAT-77, no template (briefing — solo after the workshop)

Acceptance criteria only: a magnet power-up, 5 s, radius ~4. You write the prompt, you challenge the plan, you demand the proof. Verify: coins are pulled in, the game still ends 8/8, your validator stays green.

9. Ex09_MobileBuild — iOS build from the terminal (14:55, instructor demo)

An async build from the live editor: `unity command build --target iOS ...` and `build_status`. An Xcode signing error is fine today — the exercise is the pipeline.

```
$ unity command build --target iOS --outputPath Builds/iOS-Ex09 \
  --scenes Assets/Workshop/Ex09_MobileBuild/Ex09_Start.unity --confirm true
$ unity command build_status
```

TROUBLESHOOTING

Symptom	Fix
unity status is not "ready"	The editor may be compiling; wait and retry. If it persists: check for Safe Mode with <code>unity pipeline list</code>
The agent strays outside the folder	Reject the plan; say "work only inside Assets/Workshop/ExNN_..." — CLAUDE.md already says so
eval permission/compile error	Read the console from the CLI (<code>unity command console</code>); if there is a compile error, have the agent fix it
I broke the scene / fell behind	Close without saving, open the next exercise's Start scene — you are caught up
Everything is broken	Tell the instructor: WorkshopBuilder regenerates all checkpoints (Workshop > Build All Checkpoints)

GETTING READY FOR THE CLOSING (16:40)

- ☐ Your BUG-201 loop's proof is ready: before/after eval output + a run that ends 8/8.
- ☐ Your `workshop_validate` command appears in the `unity command` list.
- ☐ The iOS build report (BuildReport summary) from the demo is at hand.
- ☐ You have written your T3 (bug resolution time) observation into the baseline metrics template.